

PART-B

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	7	0	0	07

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX} \\ S_{yX} & S_{XX} \end{pmatrix}^T$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Answer). $R^2 = \frac{SS_{\text{reg}}}{SS_T} = 1 - \frac{SS_{\text{res}}}{SS_T}$

$$SS_{\text{reg}} = \sum (y_i - \bar{y})^2 \quad SS_T = \sum (y_i - \bar{y})^2 \quad SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2$$

R^2 ea. implies the amount of variation covered by our Model.

$$r_1 = \text{CORR}(y_i, \hat{y}_i) \quad \text{X}$$

$$r_1 = \frac{\sum (y_i - \bar{y}_i)(\hat{y}_i - \bar{y}_i)}{\sqrt{\sum (y_i - \bar{y}_i)^2 \sum (\hat{y}_i - \bar{y}_i)^2}}$$

$$\text{Note :- } \bar{y}_i = \bar{\hat{y}}_i$$

$$= \frac{\sum ((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}_i))(\hat{y}_i - \bar{y}_i)}{\sqrt{\sum (y_i - \bar{y}_i)^2 \sum (\hat{y}_i - \bar{y}_i)^2}}$$

$$= \frac{\sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}_i) + \sum (\hat{y}_i - \bar{y}_i)^2}{\sqrt{\sum (y_i - \bar{y}_i)^2 \sum (\hat{y}_i - \bar{y}_i)^2}}$$

$$\text{Now } \sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}_i) = \sum e_i(\hat{y}_i) - \sum e_i \bar{y}_i = 0$$

$$\therefore r_1 = \frac{\sum (\hat{y}_i - \bar{y}_i)^2}{(\sum (y_i - \bar{y}_i)^2 \sum (\hat{y}_i - \bar{y}_i)^2)^{1/2}} = \left(\frac{\sum (\hat{y}_i - \bar{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} \right)^{1/2} = \left(\frac{SS_{\text{reg}}}{SS_{\text{res}}} \right)^{1/2}$$

$$\therefore r_1^2 = \frac{SS_{\text{reg}}}{SS_{\text{res}}} = R^2$$

~~$X^T X$~~

$\hat{y} = Hy$

$H = X(X^T X)^{-1} X^T$



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2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1' = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

Answer.) $X = \mu + LF + E$

$(n \times 1) \quad (n \times 1) \quad (n \times p)(p \times 1) \quad n \times 1$

$Cov(E, F) = 0$

$$E(E) = 0$$

~~E(E)~~

$$E(F) = 0$$

$$Var(E) = \Psi$$

$$Var(F) = I$$

$$Var(X) = LL^T + \Psi$$

Ψ is a diagonal Matrix

$$Var(X_i) = \underbrace{\sum_{j=1}^p h_{ij}^2}_{\text{Communality}} + \underbrace{\psi_i}_{\text{Specific Variance}}$$

$L \rightarrow$ Loading Matrix

$F \rightarrow$ Factor Matrix (common)

$E \rightarrow$ Specific Error

a.)

$$\tilde{L} = \begin{bmatrix} e_1' \sqrt{\lambda_1} & e_2' \sqrt{\lambda_2} \end{bmatrix}$$

where $\lambda_1' \geq \lambda_2' \geq \lambda_3' \geq \lambda_4'$

where the eigen values of S in decreasing order.

$e_1' \ e_2' \ e_3' \ e_4'$ are corresponding eigen vectors.

For our case

$$\lambda_1' = \lambda_3 \quad \lambda_2' = \lambda_2$$

$$e_1' = e_3 \quad e_2' = e_2$$

$$\tilde{L} = \begin{bmatrix} e_3 \sqrt{\lambda_3} & e_2 \sqrt{\lambda_2} \end{bmatrix}$$

$$\tilde{L} = \begin{bmatrix} e_3 \sqrt{4.2} & e_2 \sqrt{2.1} \end{bmatrix}$$

$$\tilde{L} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \\ h_{31} & h_{32} \\ h_{41} & h_{42} \end{bmatrix}$$

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b.) $\tilde{l}_1' = [0.8 \quad -0.4]$ $\therefore h_{11} = 0.8 \quad h_{12} = -0.4$

Communality for 1st variable = $(h_{11}^2 + h_{12}^2) = 0.8^2 + (-0.4)^2$
 $= 0.8$

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$$c.) \quad L = [e_3 \sqrt{\lambda_3} \quad e_2 \sqrt{\lambda_2}]$$

$$\text{let } e_3 = \begin{bmatrix} e_{31} \\ e_{32} \\ e_{33} \\ e_{34} \end{bmatrix} \quad \therefore e_2 = \begin{bmatrix} e_{21} \\ e_{22} \\ e_{23} \\ e_{24} \end{bmatrix}$$

$$L = \begin{bmatrix} e_{31} \sqrt{\lambda_3} & e_{21} \sqrt{\lambda_2} \\ e_{32} \sqrt{\lambda_3} & e_{22} \sqrt{\lambda_2} \\ e_{33} \sqrt{\lambda_3} & e_{23} \sqrt{\lambda_2} \\ e_{34} \sqrt{\lambda_3} & e_{24} \sqrt{\lambda_2} \end{bmatrix}$$

Now, we will find LL^T .

$$LL^T = \begin{bmatrix} e_{31} \sqrt{\lambda_3} & e_{21} \sqrt{\lambda_2} \\ e_{32} \sqrt{\lambda_3} & e_{22} \sqrt{\lambda_2} \\ e_{33} \sqrt{\lambda_3} & e_{23} \sqrt{\lambda_2} \\ e_{34} \sqrt{\lambda_3} & e_{24} \sqrt{\lambda_2} \end{bmatrix} \begin{bmatrix} e_{31} \sqrt{\lambda_3} & e_{32} \sqrt{\lambda_3} & e_{33} \sqrt{\lambda_3} & e_{34} \sqrt{\lambda_3} \\ e_{21} \sqrt{\lambda_2} & e_{22} \sqrt{\lambda_2} & e_{23} \sqrt{\lambda_2} & e_{24} \sqrt{\lambda_2} \end{bmatrix}$$

$4 \times 2 \quad 2 \times 4$

$$LL^T = \begin{bmatrix} (e_{31}^2 \lambda_3 + e_{21}^2 \lambda_2) & & & \\ & e_{32}^2 \lambda_3 + e_{22}^2 \lambda_2 & & \\ & & e_{33}^2 \lambda_3 + e_{23}^2 \lambda_2 & \\ & & & e_{34}^2 \lambda_3 + e_{24}^2 \lambda_2 \end{bmatrix}$$

$$\therefore \sum_{i=1}^p h_i^2 = \text{Tr}(LL^T) = e_{31}^2 \lambda_3 + e_{21}^2 \lambda_2 + e_{32}^2 \lambda_3 + e_{22}^2 \lambda_2 + e_{33}^2 \lambda_3 + e_{23}^2 \lambda_2 + e_{34}^2 \lambda_3 + e_{24}^2 \lambda_2$$

$$= \lambda_3 (e_{31}^2 + e_{32}^2 + e_{33}^2 + e_{34}^2)$$

$$+ \lambda_2 (e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2)$$

$$= \lambda_3 e_3^T e_3 + \lambda_2 e_2^T e_2$$

$$= \lambda_3 (1) + \lambda_2 (1)$$

$\therefore$ Eigen vectors given are Normalized.

$$= \lambda_3 + \lambda_2$$

Hence Proved.

04

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Answer.) $x_1, x_2, \dots, x_n$ is a Random Sample

$$\text{Mean} = \mu \in \mathbb{R} \quad \text{Var} = \mu^2$$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n x_i$$

Now, we know if $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$

then if $g(\hat{\theta})$ is a continuous function.

$$\text{s.t. } g'(\theta) \neq 0$$

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 [g'(\theta)]^2)$$

Now, for our problem.

$$\sqrt{n}(\hat{\bar{X}}_n - \bar{X}_n) \not\xrightarrow{d} N(0, \mu^2)$$

For our func. $g(\bar{X}_n)$

$$\sqrt{n}(g(\hat{\bar{X}}_n) - g(\bar{X}_n)) \not\xrightarrow{d} N(0, \sigma^2)$$

s.t. σ^2 is independent of μ .

if we take $g(\bar{X}_n) = \ln(\bar{X}_n)$

$$\text{then } \sqrt{n}(g(\hat{\bar{X}}_n) - g(\bar{X}_n)) \longrightarrow N(0, \mu^2 [g'(\bar{X}_n)]^2)$$

oo

$$\therefore \sqrt{n}(g(\hat{\bar{X}}_n) - g(\bar{X}_n)) \longrightarrow N(0, \mu^2 \frac{1}{\mu^2}) \longrightarrow N(0, 1)$$

However,

$\bar{X}_n$ can have negative value which will make our function not-defined.

$\therefore$ we can not put negative value inside

log.

$$\therefore \sqrt{n}(g(\hat{\bar{X}}_n) - g(\bar{X}_n)) \not\xrightarrow{d} N(0, \mu^2 [g'(\bar{X}_n)]^2)$$

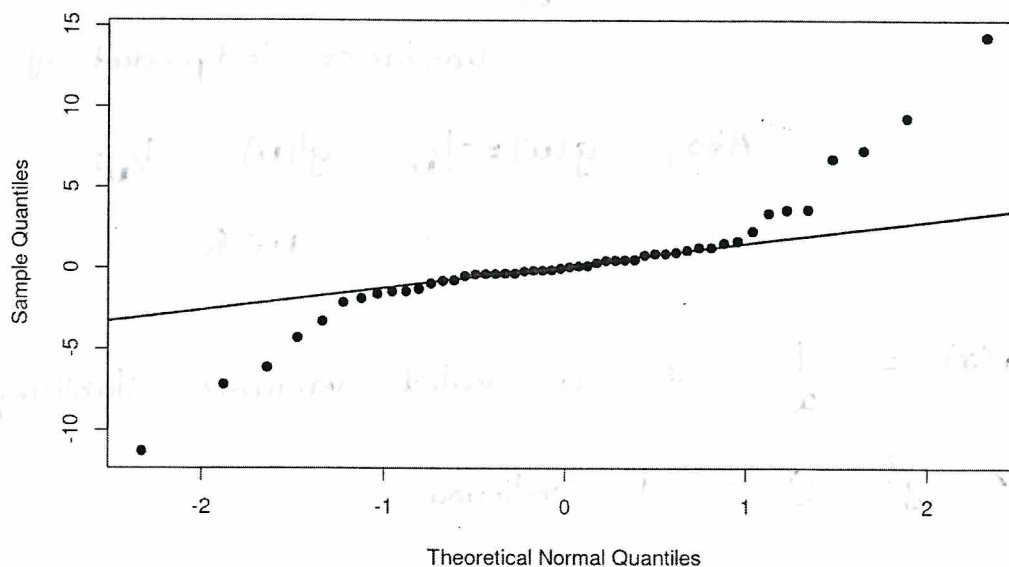
$$\sqrt{n}(g(\hat{\bar{X}}_n) - g(\bar{X}_n)) \not\xrightarrow{d} N(0, 1)$$

Continued =

Hence, we take $g(\bar{X}_n) = \frac{-1}{\bar{X}_n}$

$$\therefore g'(\bar{X}_n) = \frac{+1}{(\bar{X}_n)^2}$$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Answer.) Yes, this data has a $q-q$ plot which shows that data follows Normal Distribution.

Contradictory.

If Normality Assumption is violated then, Cauchy distribution would be more appropriate.

oo

Answer 3.)

$$\therefore \sqrt{n} (g(\bar{x}_n) - g(x_n)) \rightarrow N(0, 1)$$

$\therefore$ distribution has
variance independent of n

Also, $g(u) = -1/u$ $g'(u) = 1/u^2$
 $u \in \mathbb{R}$

Hence, $g(x) = \frac{1}{x}$ is a valid variance stabilizing transformation

$\therefore$ ~~g~~ g is a continuous

$$\therefore g(x) = \ln(|x|)$$

Now, it is a continuous function.

$$g'(x) = \left\{ \frac{1}{x} \right\} \neq$$

$$\therefore g'(u) = 1/u \neq 0$$

Hence .

~~$$\sqrt{n} (g(\bar{x}_n) - g(u)) \rightarrow N(0, u^2 \times (1/u^2))$$~~

$$\sqrt{n} (g(\bar{x}_n) - g(u)) \rightarrow N(0, u^2 \times (g'(u))^2)$$

$$\therefore \sqrt{n} (g(\bar{x}_n) - g(u)) \rightarrow N(0, u^2 \times 1/u^2) \\ \sim N(0, 1)$$

$$\therefore g(\bar{x}_n) \sim N(g(u), 1/n)$$

$$g(\bar{x}_n) \sim N(\ln|u|, 1/n)$$

Variance is independent of n .

$\therefore g(\bar{x}_n) = \log_e(|\bar{x}_n|)$ is a variance stabilizing transformation.

